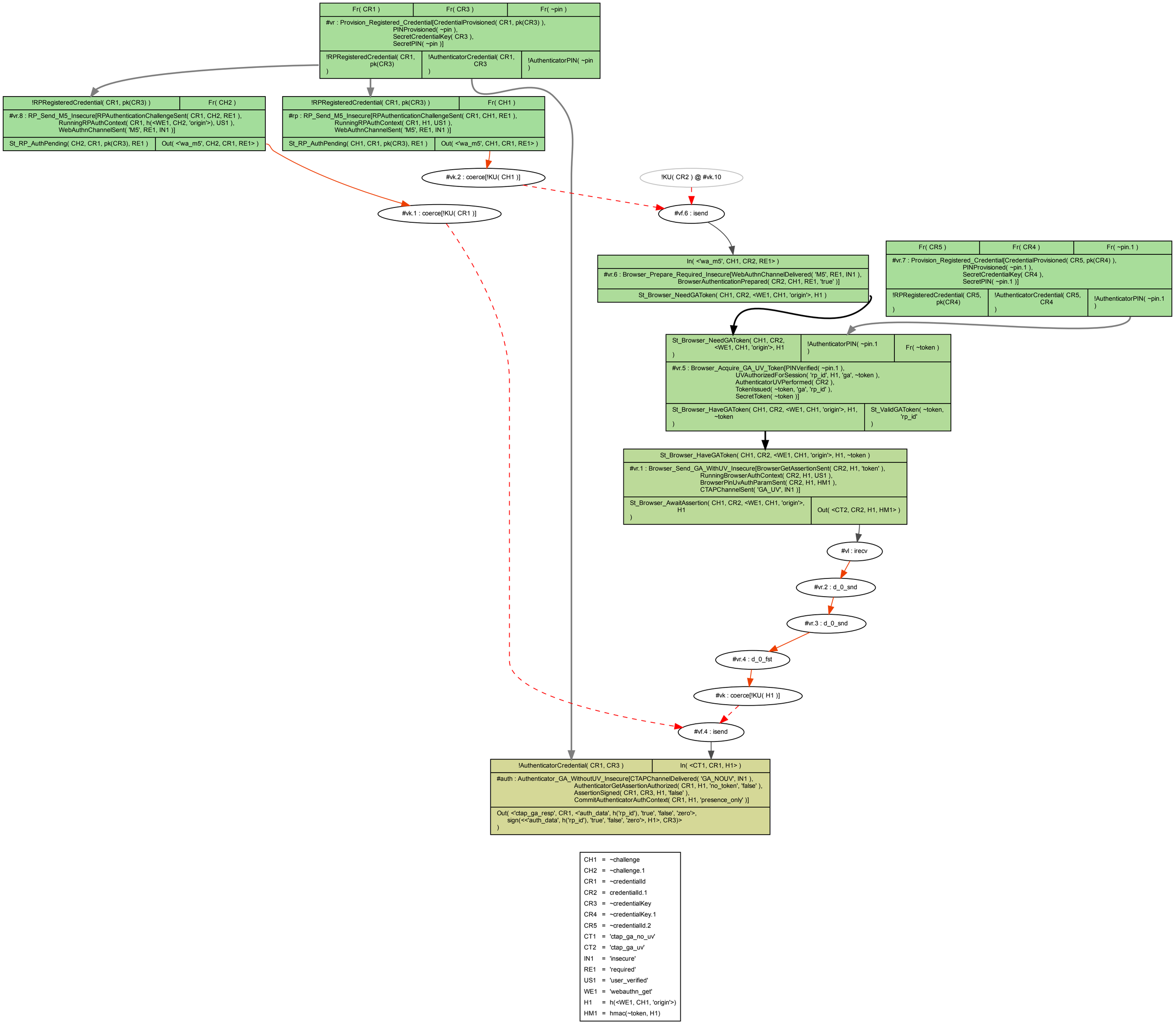
Fr(CR1) | Fr(CR3) | Fr(~pin)
#vr : Provision_Registered_Credential[CredentialProvisioned(CR1, pk(CR3)), PINProvisioned(~pin), SecretCredentialKey(CR3), SecretPIN(~pin)]
!RPRegisteredCredential(CR1, pk(CR3)) | !AuthenticatorCredential(CR1, CR3) | !AuthenticatorPIN(~pin)
!RPRegisteredCredential(CR1, pk(CR3)) | Fr(CH2)
#vr.8 : RP_Send_M5_Insecure[RPAuthenticationChallengeSent(CR1, CH2, RE1), RunningRPAuthContext(CR1, h(<WE1, CH2, 'origin'>), US1), WebAuthnChannelSent('M5', RE1, IN1)]
St_RP_AuthPending(CH2, CR1, pk(CR3), RE1) | Out(<'wa_m5', CH2, CR1, RE1>)
!RPRegisteredCredential(CR1, pk(CR3)) | Fr(CH1)
#rp : RP_Send_M5_Insecure[RPAuthenticationChallengeSent(CR1, CH1, RE1), RunningRPAuthContext(CR1, H1, US1), WebAuthnChannelSent('M5', RE1, IN1)]
St_RP_AuthPending(CH1, CR1, pk(CR3), RE1) | Out(<'wa_m5', CH1, CR1, RE1>)
#vk.2 : coerce[!KU(CH1)]
#vk.1 : coerce[!KU(CR1)]
!KU(CR2) @ #vk.10
#vf.6 : isend
In(<'wa_m5', CH1, CR2, RE1>)
#vr.6 : Browser_Prepare_Required_Insecure[WebAuthnChannelDelivered('M5', RE1, IN1), BrowserAuthenticationPrepared(CR2, CH1, RE1, 'true')]
St_Browser_NeedGAToken(CH1, CR2, <WE1, CH1, 'origin'>, H1)
Fr(CR5) | Fr(CR4) | Fr(~pin.1)
#vr.7 : Provision_Registered_Credential[CredentialProvisioned(CR5, pk(CR4)), PINProvisioned(~pin.1), SecretCredentialKey(CR4), SecretPIN(~pin.1)]
!RPRegisteredCredential(CR5, pk(CR4)) | !AuthenticatorCredential(CR5, CR4) | !AuthenticatorPIN(~pin.1)
St_Browser_NeedGAToken(CH1, CR2, <WE1, CH1, 'origin'>, H1) | !AuthenticatorPIN(~pin.1) | Fr(~token)
#vr.5 : Browser_Acquire_GA_UV_Token[PINVerified(~pin.1), UVAuthorizedForSession('rp_id', H1, 'ga', ~token), AuthenticatorUVPerformed(CR2), TokenIssued(~token, 'ga', 'rp_id'), SecretToken(~token)]
St_Browser_HaveGAToken(CH1, CR2, <WE1, CH1, 'origin'>, H1, ~token) | St_ValidGAToken(~token, 'rp_id')
St_Browser_HaveGAToken(CH1, CR2, <WE1, CH1, 'origin'>, H1, ~token)
#vr.1 : Browser_Send_GA_WithUV_Insecure[BrowserGetAssertionSent(CR2, H1, 'token'), RunningBrowserAuthContext(CR2, H1, US1), BrowserPinUvAuthParamSent(CR2, H1, HM1), CTAPChannelSent('GA_UV', IN1)]
St_Browser_AwaitAssertion(CH1, CR2, <WE1, CH1, 'origin'>, H1) | Out(<CT2, CR2, H1, HM1>)
#vl : irecv
#vr.2 : d_0_snd
#vr.3 : d_0_snd
#vr.4 : d_0_fst
#vk : coerce[!KU(H1)]
#vf.4 : isend
!AuthenticatorCredential(CR1, CR3) | In(<CT1, CR1, H1>)
#auth : Authenticator_GA_WithoutUV_Insecure[CTAPChannelDelivered('GA_NOUV', IN1), AuthenticatorGetAssertionAuthorized(CR1, H1, 'no_token', 'false'), AssertionSigned(CR1, CR3, H1, 'false'), CommitAuthenticatorAuthContext(CR1, H1, 'presence_only')]
Out(<'ctap_ga_resp', CR1, <'auth_data', h('rp_id'), 'true', 'false', 'zero'>, sign(<<'auth_data', h('rp_id'), 'true', 'false', 'zero'>, H1>, CR3)>)
CH1 = ~challenge
CH2 = ~challenge.1
CR1 = ~credentialId
CR2 = credentialId.1
CR3 = ~credentialKey
CR4 = ~credentialKey.1
CR5 = ~credentialId.2
CT1 = 'ctap_ga_no_uv'
CT2 = 'ctap_ga_uv'
IN1 = 'insecure'
RE1 = 'required'
US1 = 'user_verified'
WE1 = 'webauthn_get'
H1 = h(<WE1, CH1, 'origin'>)
HM1 = hmac(~token, H1)